



Mixed-Type Data Clustering

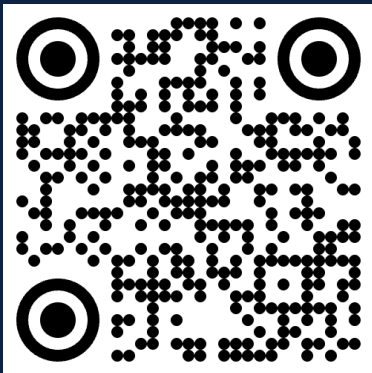
The Advanced R Series: Clustering & Classification

Slides and code created by Jesse Ghashti

March 11, 2026

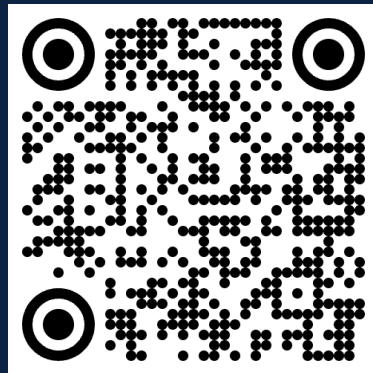
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Check out our other
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GitHub code and
slides for today's
workshop (and pre-
vious workshops)



Alternatively, code/slides available at the bottom of
<https://csc-ubc-okanagan.github.io/workshops/>

Workshop Series Overview



Session	Topic	Date/Time
1	Hierarchical Models	Jan 14, 12:00 PM
2	Centroid-Based Models	Jan 19, 3:30 PM
3	Fuzzy Clustering	Jan 28, 1:00 PM
4	Distribution-Based Models	Feb 2, 3:30 PM
5	Density-Based Models	Feb 9, 3:30 PM
6	Graph-Based Models	Feb 23, 3:30 PM
7	Mixed-Type Data Quantification	Mar 4, 1:00 PM
8	Mixed-Type Data Clustering	Mar 11, 1:00 PM
9	Bias Reduction and Fairness	Mar 18, 1:00 PM
10	Dimensionality Reduction 1 of 2	Mar 23, 3:30 PM
11	Dimensionality Reduction 2 of 2	Mar 30, 3:30 PM

What we learned...

- Discussed the challenges of mixed-type data (continuous + categorical)
- Learned Gower's distance: range-normalized continuous + simple matching categorical
- Explored variants: Podani, Wishart, Huang distances
- Introduced weighted dissimilarity measures
- Compared geographic distances: Haversine vs Euclidean
- Noted that optimal weighting is an active research area

Today we will...

- Review Gower's distance as our dissimilarity.
- Learn PAM (Partitioning Around Medoids) for dissimilarity-based clustering
- Introduce K -Prototypes for mixed-type centroid-based clustering
- Apply hierarchical clustering with Gower distance
- Revisit spectral clustering (Workshop 6) with mixed-type similarity
- Compare methods on simulated workforce data and real health records

Today we require...

```
library('cluster')      # daisy, pam, silhouette
library('clustMixType') # kproto
library('dendextend')   # colored dendrograms
library('ggplot2')      # visualization
library('dplyr')        # data manipulation
library('gridExtra')    # plot arrangement
library('viridis')      # color palettes
```

From Workshop 7

Gower's distance computes pairwise dissimilarity for mixed-type data:

$$d_G(i, j) = \frac{\sum_{k=1}^p w_k \cdot d_k(i, j)}{\sum_{k=1}^p w_k}$$

Variable-specific distances

- Continuous: $d_k(i, j) = \frac{|x_{ik} - x_{jk}|}{\text{range}_k}$ (range-normalized)
- Nominal: $d_k(i, j) = \mathbf{1}[x_{ik} \neq x_{jk}]$ (0 if same, 1 if different)
- Ordinal: Ranked then treated as continuous
- Binary: Depends on symmetric vs asymmetric treatment

Gower gives us a distance matrix $\in [0, 1]$. How do we cluster with it?

Equal Weights Today

In Workshop 7, we discussed that Gower's distance can incorporate variable weights w_k

Today we use equal weights ($w_k = 1$ for all k).

Why? Optimal weight selection is a complex topic that could be an entire workshop series on its own.

- Variable weighting often requires domain expertise (for which I do not have on this data) to determine variable importance; or
- We need a method to determine variable important, for which there are many (Inverse variance weighting, Correlation-based adjustments, Nonparametric Approaches, Optimization techniques, ...)

Measuring Clustering Quality

Accuracy (with label matching): Proportion of correctly assigned observations after optimal permutation of cluster labels.

Adjusted Rand Index (ARI):

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]} \in [-1, 1]$$

Measures agreement between two partitions, adjusted for chance. $\text{ARI} = 1$ is perfect agreement.

Information-Theoretic Cluster Evaluation

Mutual Information measures shared information between partitions:

$$MI(U, V) = \sum_{i=1}^{|U|} \sum_{j=1}^{|V|} P(i, j) \log \frac{P(i, j)}{P(i)P(j)}$$

Normalized Mutual Information:

$$NMI(U, V) = \frac{2 \cdot MI(U, V)}{H(U) + H(V)} \in [0, 1]$$

where $H(\cdot)$ is a function determining the entropy of a clustering.

- $NMI = 0$: Independent partitions (no shared information)
- $NMI = 1$: Perfect agreement (identical partitions)
- Unlike ARI, NMI is always non-negative

Visualizing Dissimilarity Matrices

Problem: Mixed-type data has no natural coordinate system for plotting.

Possible Solution: Multidimensional Scaling (MDS) finds a low-dimensional embedding that preserves pairwise distances.

Classical MDS (Principal Coordinates Analysis)

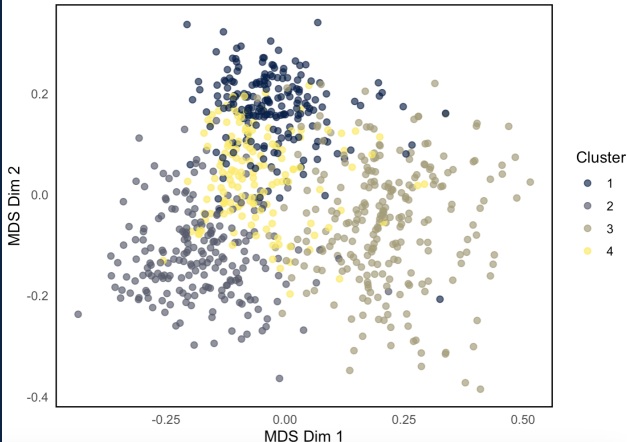
1. Start with $n \times n$ distance matrix D
2. Double-center: $B = -\frac{1}{2}HD^{(2)}H$ where $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$
3. Eigendecompose $B = V\Lambda V^T$ (Sound familiar?)
4. Take top k eigenvectors scaled by $\sqrt{\lambda}$

Result: Coordinates in $\mathbb{R}^k$ where Euclidean distances approximate original Gower distances.

```
mixedDf <- readRDS("simulatedMixedData.RDS")
mixedForClust <- mixedDf %>% select(-trueCluster)
gowerDist <- daisy(mixedForClust, metric = "gower")
mdsCoords <- cmdscale(gowerDist, k = 2)
mixedDf$mds1 <- mdsCoords[, 1]
mixedDf$mds2 <- mdsCoords[, 2]
```

MDS gives us a 2D representation of mixed-type observations, so we get a scatter plot visualization of clusters.

True Clusters



Mixed-Type Dataset: $n = 800$ observations

Continuous variables:

- age: 20–68 years
- income: \$18k–\$350k
- yearsExperience: 0–45 years

Categorical variables:

- education: HighSchool, Bachelors, Masters, PhD (ordinal)
- region: North, South, East, West (nominal)
- sector: Tech, Finance, Healthcare, Retail, Education (nominal)
- satisfaction: 1–5 (ordinal)
- employed: TRUE/FALSE (binary)

```
mixedDf <- readRDS("simulatedMixedData.RDS")  
head(mixedDf)
```

Throughout our workshops, we've addressed K -means limitations, why stop now?

K -means limitation # 999: Centroids are *means* of feature vectors — not defined for categorical data.

K -means limitation # 1000: K -means cannot handle categorical data so we'd have to use homogenization techniques Workshop 7 (e.g., dummy-coding, treating categorical as continuous, etc.).

PAM solution uses **medoids** instead of centroids.

- A **medoid** is an actual observation that minimizes total dissimilarity to all other points in its cluster
- Works directly with any dissimilarity matrix (including Gower!)
- Generally shown to be more robust to outliers than K -means

Algorithm

1. Initialize k medoids
2. Assign each point to nearest medoid
3. For each medoid, consider swapping with non-medoid (SWAP phase)
4. Accept swap if total dissimilarity decreases
5. Repeat until no improvement

Unlike K -means centroids (which may not correspond to any real data point), PAM medoids are **actual observations** from the dataset.

Advantages

- Directly interpretable: “This patient represents Cluster 2”
- No need to handle mixed types in centre calculation

Disadvantages:

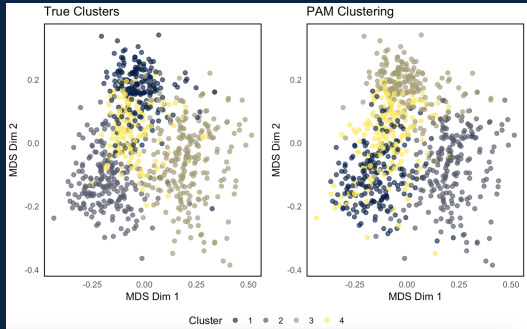
- More complexity per iteration (slower than K -means)
- Medoid may be atypical if cluster has substructure

```
pamResult4 <- pam(gowerDist, k = 4, diss = TRUE)
mixedDf$pamCluster <- factor(pamResult4$clustering)
mixedForClust[pamResult4$medoids, ]
```

Medoid observations:

Cl	Age	Income	Education	Sector
1	47	189k	Masters	Finance
2	43	33k	HS	Retail
3	28	111k	Bach	Tech
4	44	75k	Masters	Education

Cl	Exp.	Employed	Region	Satis.
1	23	North	True	3
2	23	South	True	2
3	6	West	True	3
4	13	East	True	4

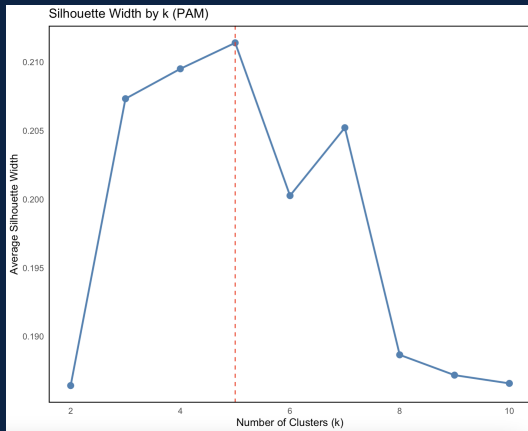


Choosing K : Silhouette Analysis

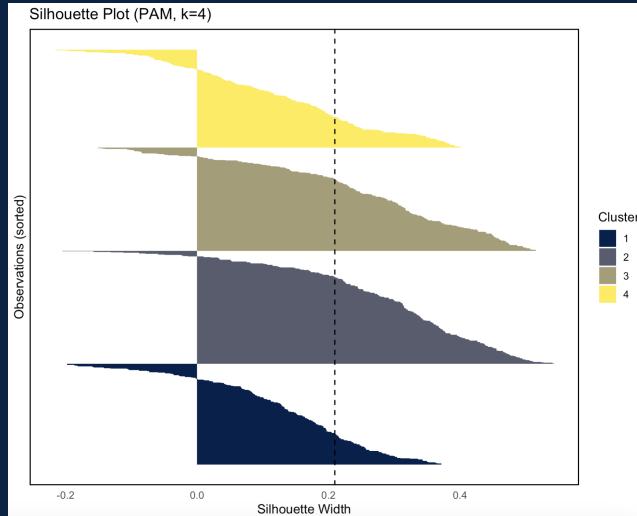


```
silWidths <- numeric(8)
for (k in 2:9) {
  pamK <- pam(gowerDist, k = k, diss = TRUE)
  silWidths[k-1] <- pamK$silinfo$avg.width
}
silDf <- data.frame(k = 2:9, silhouette = silWidths)
optimalK <- silDf$k[which.max(silDf$silhouette)]
```

Silhouette width measures how similar each point is to its own cluster vs other clusters. Higher is better.



Silhouette Plot



K -Prototypes (Huang 1998) extends K -means to mixed-type data by defining a **prototype** as

- Mean for continuous variables
- Mode for categorical variables

Distance function is

$$d(x_i, Q_l) = \sum_{j \in \text{cont}} (x_{ij} - q_{lj})^2 + \lambda \sum_{j \in \text{cat}} \mathbf{1}[x_{ij} \neq q_{lj}]$$

where λ **parameter** controls trade-off between continuous and categorical contributions. The default is $\lambda = (\text{average std of continuous vars})$. Q_l is the prototype vector of the l th cluster with entries q_{lj}

Algorithm works just like K -means — assign to nearest prototype, update prototypes, repeat.

Fitting K -Prototypes

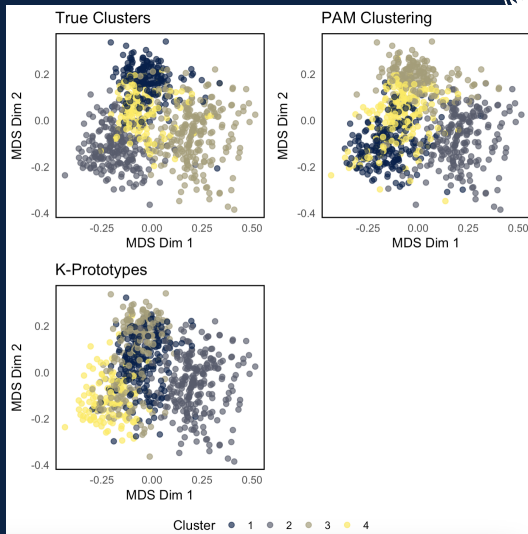


```
kprotoResult <- kproto(mixedForClust, k = 4,  
                        nstart = 25, verbose = FALSE)  
mixedDf$kprotoCluster <- factor(kprotoResult$cluster)  
kprotoResult$centers
```

Prototype centres:

Cl	Age	Income	Edu	Sector
1	39.9	82.8k	PhD	Education
2	38.4	39.6k	HS	Retail
3	37.9	141.4k	Masters	Tech
4	51.1	271.4k	Masters	Finance

Cl	Exp.	Employed	Region	Satis.
1	14.2	North	True	3
2	18.0	South	FALSE	1
3	14.9	West	True	4
4	27.1	East	True	3



	PAM	K -Prototypes
Input	Dissimilarity matrix	Raw data
Center type	Medoid (actual obs)	Prototype (mean/mode)
Distance	Any (e.g., Gower)	Squared Euclidean + matching
Interpretability	High (real observation)	Moderate (synthetic centre)
Flexibility	High	Moderate

PAM is more flexible (works with any distance) but slower. K -Prototypes is faster but uses a specific distance formulation.

Revisiting Workshop 1

Recall that hierarchical clustering builds a tree (dendrogram) by iteratively merging or splitting clusters.

Agglomerative:

1. Start with n singleton clusters
2. Merge closest pair according to linkage criterion
3. Repeat until one cluster remains
4. Cut dendrogram at desired height to get k clusters

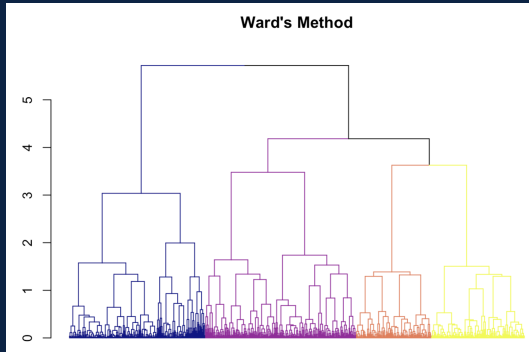
Linkage methods (amongst numerous others)

- **Complete**-> Max distance between cluster members
- **Average**-> Mean distance between all pairs
- **Ward**-> Minimize within-cluster variance

Fitting Hierarchical Clustering



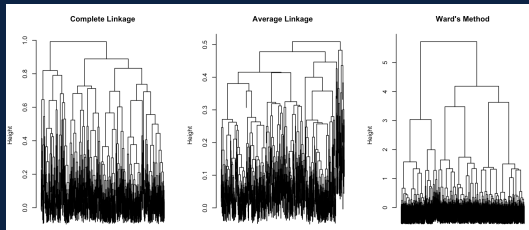
```
hclustWard <- hclust(gowerDist, method = "ward.D2")
mixedDf$hclustCluster <- factor(cutree(hclustWard, k = 4))
dendWard <- as.dendrogram(hclustWard)
dendWard <- color_branches(dendWard, k = 4,
  col = viridis(4, option = "cividis"))
plot(dendWard, leaflab = "none")
```



Comparing Linkage Methods



Linkage	Accuracy	ARI
Complete	0.494	0.259
Average	0.494	0.208
Ward	0.735	0.417



Ward's method outperforms complete and average linkage on this data. Ward tends to produce more compact, spherical clusters.

Revisiting Workshop 6

Recall that spectral clustering works on a **similarity graph**:

1. Build similarity matrix W from data
2. Compute normalized graph Laplacian $L_{\text{sym}} = I - D^{-1/2}WD^{-1/2}$
3. Extract bottom k eigenvectors (smallest eigenvalues)
4. Row-normalize and apply K -means

For mixed-type data we can convert Gower distance to similarity as

$$W_{ij} = 1 - d_G(i, j) \quad (\text{linear})$$

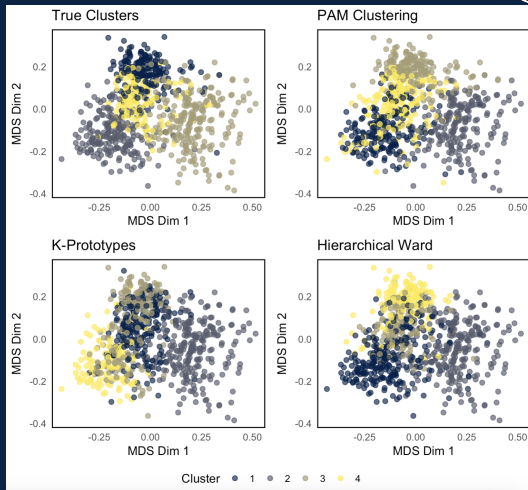
or

$$W_{ij} = \exp\left(-\frac{d_G(i, j)^2}{2\sigma^2}\right) \quad (\text{Gaussian kernel})$$

Fitting Spectral Clustering



```
gowerMat <- as.matrix(gowerDist)
sigma <- 0.26
gowerSim <- exp(-gowerMat^2 / (2 * sigma^2))
diag(gowerSim) <- 0
dVec <- rowSums(gowerSim)
dInvSqrt <- diag(1 / sqrt(pmax(dVec, 1e-10)))
Lsym <- diag(nrow(gowerMat)) -
  dInvSqrt %*% gowerSim %*% dInvSqrt
eigenResult <- eigen(Lsym, symmetric = TRUE)
U <- eigenResult$vectors[, (n-3):n]
rn <- sqrt(rowSums(U^2)); rn[rn == 0] <- 1
T <- U / rn
spectralClust <- kmeans(T, centers = 4, nstart = 50)
```



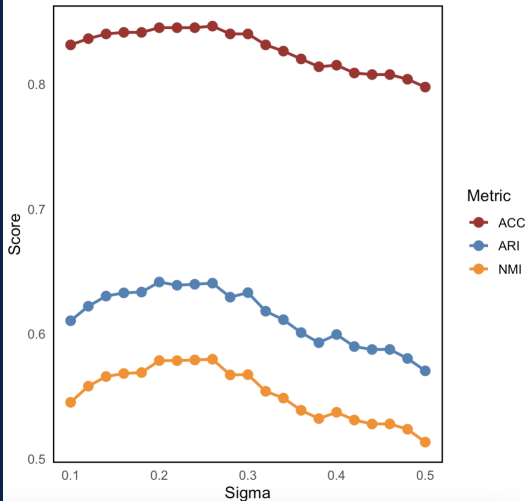
Tuning the Bandwidth σ



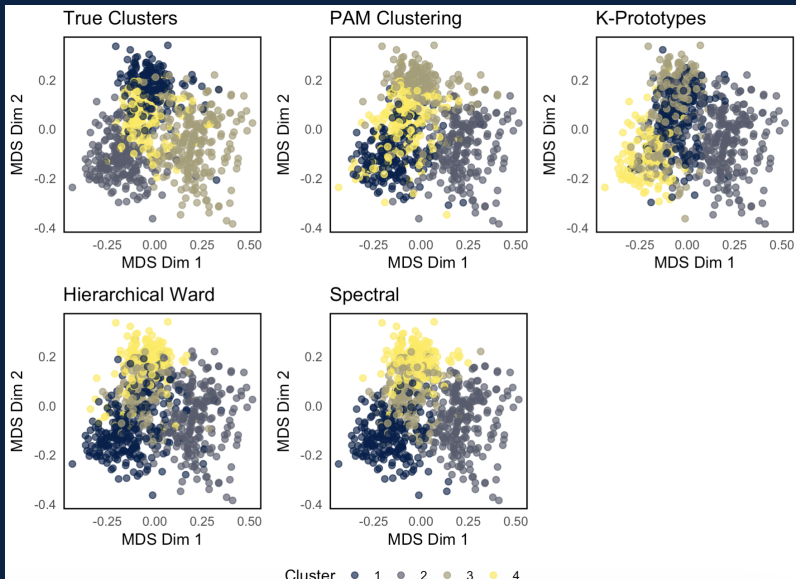
```
sigmaVals <- seq(0.1, 0.5, by = 0.02)
sigmaResults <- data.frame(
  sigma = sigmaVals, ACC = NA, ARI = NA, NMI = NA)
for (i in seq_along(sigmaVals)) {
  s <- sigmaVals[i]
  gowerSim <- exp(-gowerMat^2 / (2 * s^2))
  diag(gowerSim) <- 0
  #run the rest of spectral clustering code
  sigmaResults$ACC[i] <- computeACC(...)
}
```

Best $\sigma = 0.26$ with a maximum accuracy of 0.846.

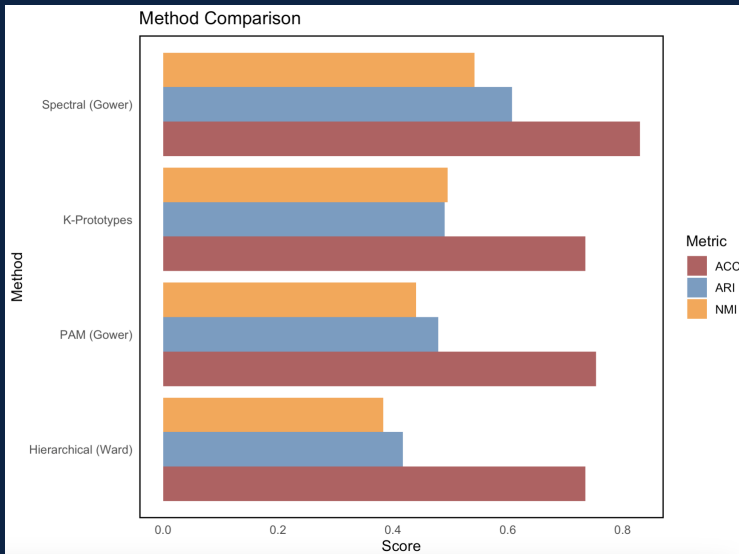
Spectral Clustering: Effect of Sigma



Four-Way Method Comparison



Performance Summary of Simulated Data



Method Agreement



Pairwise ACC Between Methods

How similar are the partitions produced by different methods?

Agreement Between Methods (ACC)



Cluster Profiles (Spectral)



Cluster	n	Age	Income	Exp	%MA/PhD	%Empl	Top Sector
1	183	30.5	126k	8.2	50.3	95.1	Tech
2	223	50.2	199k	26.4	74.9	89.2	Finance
3	164	40.6	81k	14.2	81.1	89.0	Education
4	230	38.8	40k	18.5	10.9	74.3	Retail

Clusters reveal some useful information about the workforce: young tech workers, senior professionals, educators, and service workers.

Real Mixed-Type Health Records

Dataset is the Cleveland Heart Disease (from UCI Repository)

- $n = 297$ patients (after removing missing values)
- 5 continuous: age, resting BP, cholesterol, max HR, ST depression
- 4 binary: sex, fasting blood sugar, exercise angina, ...
- 4 categorical: chest pain type, resting ECG, slope, thalassemia

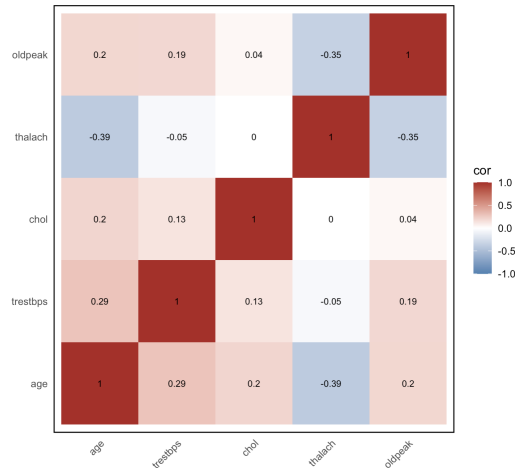
We want to cluster patients based on clinical features, then examine correspondence with diagnosis (Disease vs No Disease). We ask... can unsupervised clustering identify meaningful patient subgroups?

Heart Disease: Continuous Variables



Variable	Mean	Min	Max
Age (years)	54.5	29	77
Resting BP (mmHg)	131.7	94	200
Cholesterol (mg/dl)	247.4	126	564
Max Heart Rate	149.6	71	202
ST Depression	1.1	0	6.2

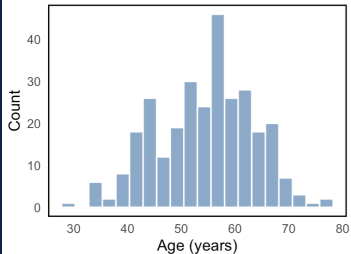
Correlation of Continuous Variables



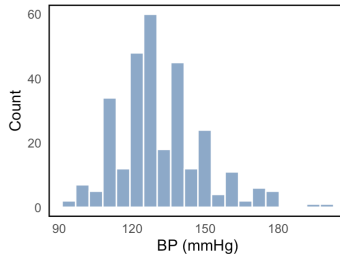
Heart Disease: Exploratory Analysis



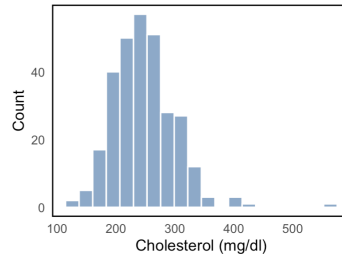
Age



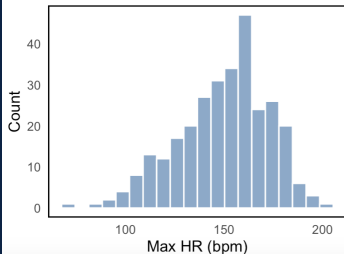
Resting Blood Pressure



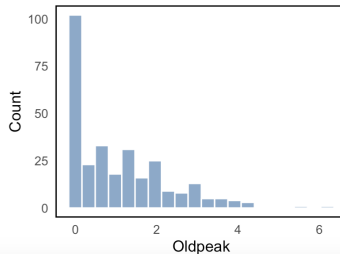
Cholesterol



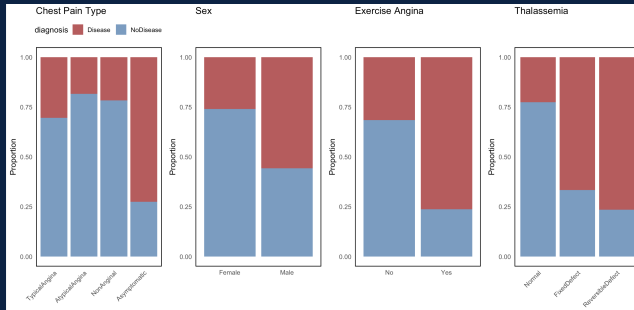
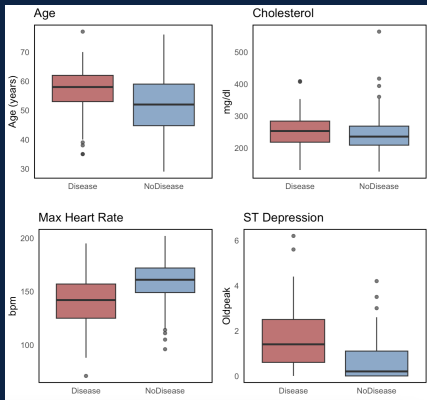
Max Heart Rate



ST Depression



Heart Disease: Exploratory Analysis

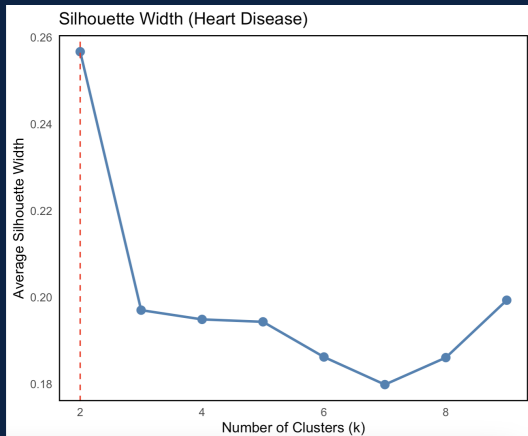


Clustering Heart Disease Patients

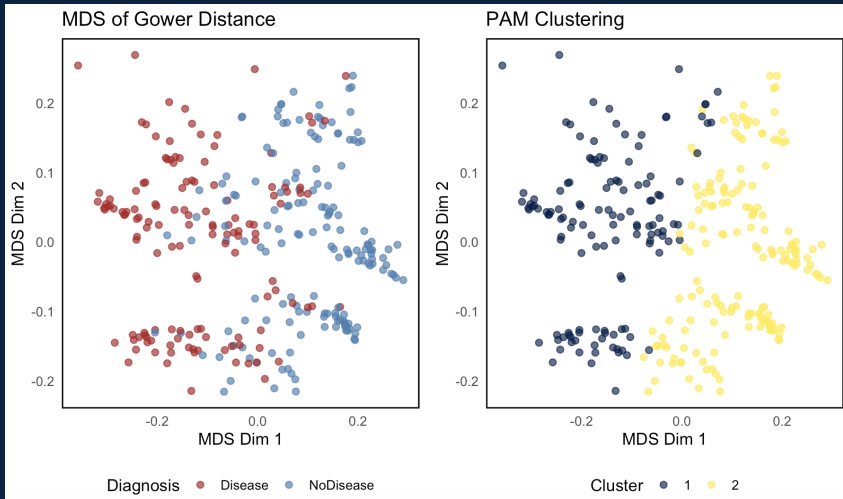


```
heartForClust <- heartClean %>%  
  select(-num, -diagnosis)  
gowerHeart <- daisy(heartForClust, metric = "gower")  
mdsHeart <- cmdscale(gowerHeart, k = 2)  
silWidthsHeart <- numeric(8)  
for (k in 2:9) {  
  pamK <- pam(gowerHeart, k = k, diss = TRUE)  
  silWidthsHeart[k-1] <- pamK$silinfo$avg.width  
}
```

Silhouette analysis suggests $k = 2$ clusters, aligning with the binary Disease/No Disease outcome.



Gower's Distance & Pam Clustering



All Clusterings

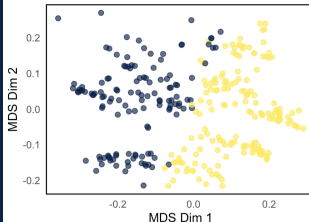


```
gowerMatHeart <- as.matrix(gowerHeart)
sigma <- 0.3
gowerSimHeart <- exp(-gowerMatHeart^2 / (2*sigma^2))
diag(gowerSimHeart) <- 0

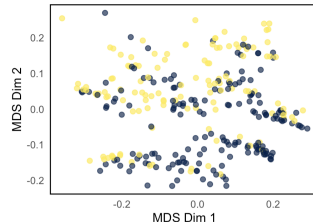
dVecHeart <- rowSums(gowerSimHeart)
dInvSqrtHeart <- diag(1/sqrt(pmax(dVecHeart, 1e-10)))
LsymHeart <- diag(nrow(gowerMatHeart)) -
  dInvSqrtHeart %*% gowerSimHeart %*% dInvSqrtHeart

eigenHeart <- eigen(LsymHeart, symmetric = TRUE)
```

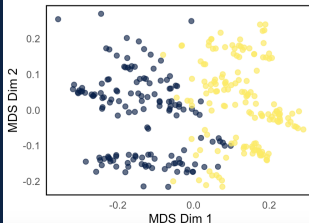
PAM Clustering



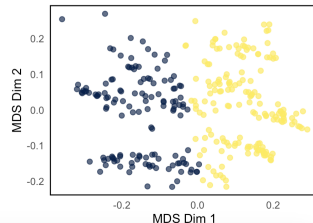
K-Prototypes



Hierarchical Ward



Spectral



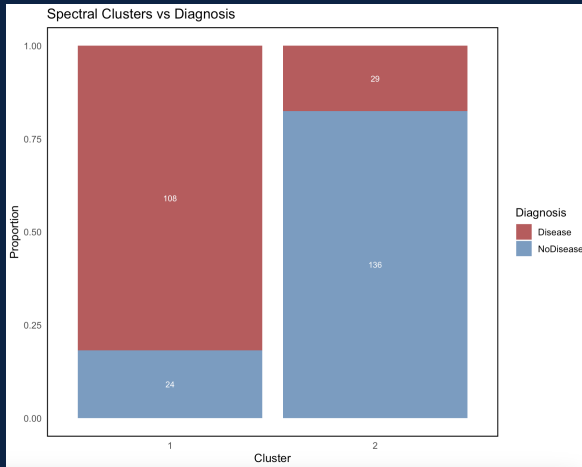
Heart Disease: Cluster vs Diagnosis



Method Accuracy vs Diagnosis

Method	Accuracy
PAM	0.808
K-Prototypes	0.579
Hierarchical	0.798
Spectral	0.822

Spectral clustering achieves 82.2% accuracy in distinguishing Disease from No Disease — using only clinical features, no labels.



Cl	n	Age	BP	MaxHR	Oldpeak	%Male	%Exang	%Disease
1	165	52.9	130	158	0.58	50.9	9.1	17.6
2	132	56.5	134	139	1.64	88.6	62.1	81.8

Cluster 2 (high-risk) characteristics:

- Older, predominantly male (88.6%)
- Lower max heart rate (139 vs 158 bpm)
- Higher ST depression (1.64 vs 0.58)
- Much higher exercise-induced angina (62.1% vs 9.1%)

Categorical Variable Profiles

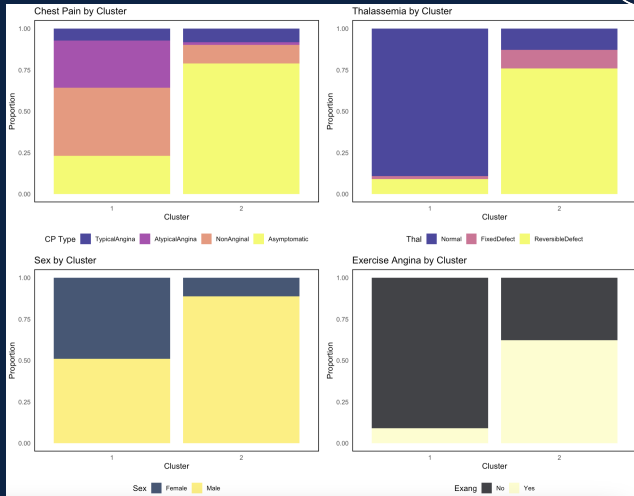


Chest Pain Type (%)

	Cl 1	Cl 2
Typical	7.3	8.3
Atypical	28.5	1.5
NonAnginal	41.2	11.4
Asymptomatic	23.0	78.8

Thalassemia (%)

	Cl 1	Cl 2
Normal	89.1	12.9
Fixed	1.8	11.4
Reversible	9.1	75.8



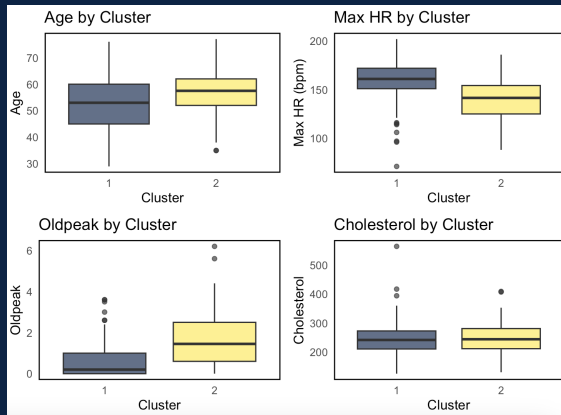
What the Clusters Reveal

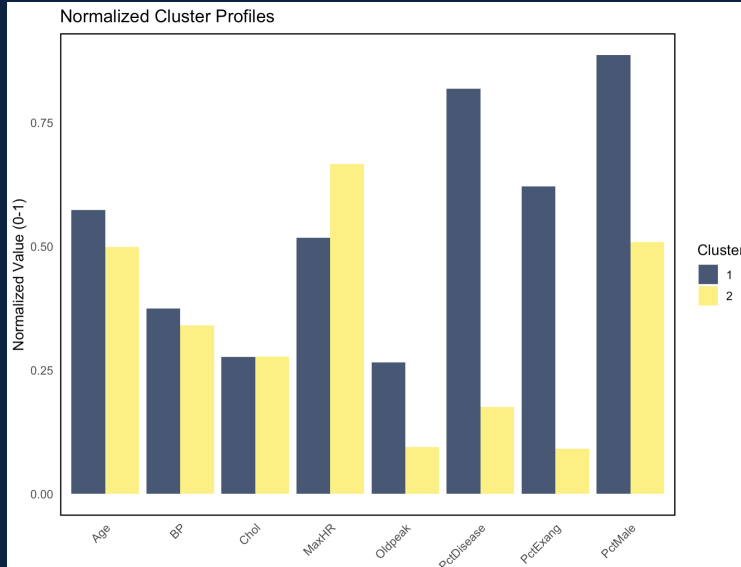
Cluster 1 (Low Risk):

- Better exercise capacity ($\uparrow$ max HR)
- Normal thallium scans (89%)
- Symptomatic chest pain (atypical/non-anginal)
- $\downarrow$ exercise-induced angina (9%)

Cluster 2 (High Risk):

- Reduced exercise capacity ($\downarrow$ max HR)
- Abnormal thallium scans (76% reversible defect)
- High proportion *asymptomatic* chest pain (79%)
- $\uparrow$ exercise-induced angina (62%)





- Gower's distance is a good start for mixed-type clustering
- PAM uses medoids (actual observations) and works with any dissimilarity matrix
- K -Prototypes combines K -means with mode-based categorical matching
- MDS projects dissimilarity matrices to visualizable coordinates
- Spectral clustering with Gaussian kernel similarity often outperforms other methods
- Different methods may capture different aspects of cluster structure — compare, compare, compare!

Additional Questions?
Book an Appointment!



Next Workshop: Mar 18, 1:00 PM

Bias Reduction and Fairness

- > So far, we've focused on clustering accuracy.
- > But what if clusters systematically disadvantage certain groups?
- > Next workshop explores fairness constraints in clustering and classification.

Thank You!

Questions?

Workshop Materials:

<https://csc-ubc-okanagan.github.io/workshops/>

Contact:

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